

Searching Techniques

Searching

- Searching is an operation which finds the location of a given element in a list.
- The search is said to be successful or unsuccessful depending on whether the element that is to be searched is found or not.
- Search Techniques
 - **Linear Search**
 - **Binary Search**

Linear Search

Linear Search

- This is the simplest method of searching. In this method, the element to be found is ***sequentially searched*** in the list. This method can be used for a sorted or an unsorted list.
- In case of sorted list (assuming in ascending order) searching starts from the 0th element and continues until the element is found or an element whose value is greater than the value being searched is reached.
- In case of unsorted list searching starts from 0th element and continues until the element is found or the end of list is reached.

Algorithm

Linear Search

Input: List of Array elements , search element, size of array

Output: Element found or Not found

Method:

LINEAR_SEARCH (ARRAY,DATA, MAXSIZE)

Index =0

Found_Flag=FALSE

WHILE Index < MAXSIZE and Found_Flag=FALSE

 IF ARRAY(Index) = DATA

 Found_Flag=TRUE

 ELSE

 Index=Index+1

 END-IF

END-WHILE

IF Found_Flag=TRUE

 PRINT "data is found"

ELSE

 PRINT "data is not found"

ENDIF

EXIT

Analysis

- best-case : $O(1)$
- average : $O(n)$.
- worst-case : $O(n)$.

Binary Search

Binary Search

- Binary search method is very fast and efficient. This method **requires that the list of elements be in sorted order.**

In this method

- To search an element we compare it with the element present at the center of the list. If it matches then the search is successful.
- Otherwise , the list is divided into two halves:
 - One from 0th element to the center element (first half)
 - Another from center element to the last element (second half)
- The searching will now proceed in either of the two halves depending upon whether the element is greater or smaller than the center element.
- If the **element is smaller than the center element** then the searching will be done in the first half, otherwise in the second half.

Binary Search Algorithm

Input: List of Array elements , search element, size of array

Output: Element found or Not found

Method:

BINARY_SEARCH(ARRAY, DATA, MAXSIZE)

Lower = 0

Upper=MAXSIZE-1

Found_Flag=FALSE

WHILE Lower <= Upper and Found_Flag=FALSE

 Mid = (Lower+Upper)/2

 IF ARRAY[Mid] = DATA

 Found_Flag=TRUE

 ELSE

 IF DATA < ARRAY[Mid]

 Upper = Mid -1

 ELSE

 Lower = Mid + 1

 END-IF

 END-IF

END-WHILE

 IF Found_Flag = TRUE

 PRINT "data is found"

 ELSE

 PRINT "data is not found"

 END-IF

EXIT

Analysis

- best-case : $O(1)$
- average : $O(\log n)$.
- worst-case : $O(\log n)$.

Limitations

- Binary search can be applied **iff** items are sorted.
- Binary search can only be applied to data structures which allow direct access to elements.

eg. we can not apply binary search to Linked List.